

Inside an atom

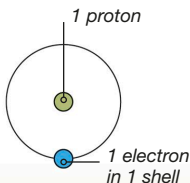
An atom is the smallest unit of an element. Every element has atoms made of a unique combination of even smaller particles, known as subatomic particles. These are the electrons, protons, and neutrons.

Subatomic particles

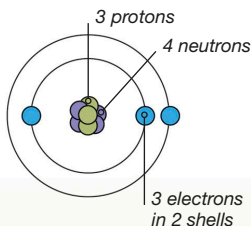
In an atom, protons and neutrons are located in its core, or nucleus. Changes to these particles are called nuclear reactions. The electrons, located outside of the nucleus, participate in chemical reactions, helping elements combine to form compounds.

Atomic number

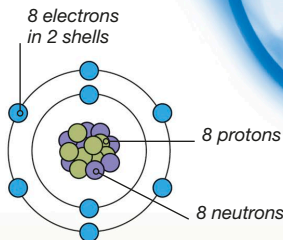
An element's atomic number is defined by the number of protons in one of its atoms. Every element has a unique atomic number. Hydrogen has the lowest atomic number (1) among the elements as it has only one proton. The number of electrons in an element always matches the atomic number.



Hydrogen atom
(atomic number 1)



Lithium atom
(atomic number 3)



Oxygen atom
(atomic number 8)

